

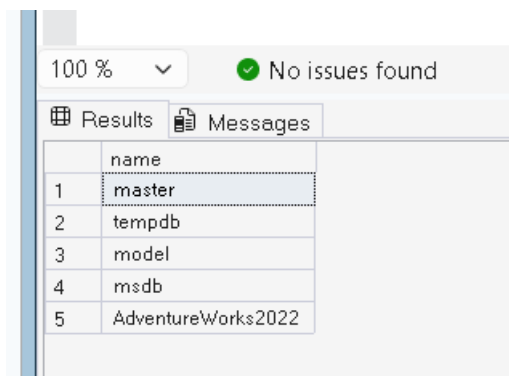
Querying SQL Server

1) Browse different system databases as well as user database.

SELECT name

FROM sys.databases;

Output



The screenshot shows a SQL Server query results window. At the top, there is a zoom level of 100% and a status bar indicating 'No issues found'. Below this, there are two tabs: 'Results' and 'Messages'. The 'Results' tab is active, displaying a table with two columns: 'name' and an implicit index column. The table contains five rows of data representing system databases.

	name
1	master
2	tempdb
3	model
4	msdb
5	AdventureWorks2022

2) Write a SELECT statement that lists the customers along with their ID numbers. Include the last names, first names, and company names.

SELECT

c.CustomerID,

p.LastName,

p.FirstName,

s.Name AS CompanyName

```

FROM Sales.Customer c
LEFT JOIN Person.Person p
ON c.PersonID = p.BusinessEntityID
LEFT JOIN Sales.Store s
ON c.StoreID = s.BusinessEntityID;

```

Output

100 %

No issues found

Results

Messages

	CustomerID	FirstName	LastName	CompanyName
1	29485	Catherine	Abel	Professional Sales and Service
2	29487	Humberto	Acevedo	The Bike Mechanics
3	29488	Pilar	Ackerman	Nationwide Supply
4	28866	Aaron	Adams	NULL
5	13323	Adam	Adams	NULL

3) Write a query that displays all the employees listed in the HumanResources.Employee table who have the job title Research and Development Engineer. Display the BusinessEntityID, LoginID, and JobTitle.

```

SELECT
    BusinessEntityID,
    LoginID,
    JobTitle
FROM HumanResources.Employee

```

WHERE JobTitle = 'Research and Development Engineer';

Output

100 %

✔ No issues found

Results

Messages

	BusinessEntity...	LoginID	JobTitle
1	8	adventure-works\diane1	Research and Development Engineer
2	9	adventure-works\gigi0	Research and Development Engineer

4) Write a query displaying all the columns of the Production.ProductCostHistory table from the rows that were modified on June 17, 2003.

```
SELECT *  
FROM Production.ProductCostHistory  
WHERE ModifiedDate = '2003-06-17';
```

Output

ProductID	Start Date	EndD ate	Standard Cost	Modified Date
-----------	---------------	-------------	------------------	------------------

No rows
returned
(AdventureWor
ks2022)

5) Rewrite the query written in Question 3, changing it so that employees who do not have the title Research and Development Engineer are displayed.

```
SELECT  
  
    BusinessEntityID,  
  
    LoginID,  
  
    JobTitle  
  
FROM HumanResources.Employee  
  
WHERE JobTitle <> 'Research and Development Engineer';
```

Output



The screenshot shows a SQL Server Enterprise Manager interface. At the top, there is a status bar with '100 %' and 'No issues found'. Below it, there are tabs for 'Results' and 'Messages'. The 'Results' tab is active, displaying a table with 4 columns: 'BusinessEntityID', 'LoginID', and 'JobTitle'. The table contains 8 rows of data. The first row has '1' for BusinessEntityID, 'adventure-works\ken0' for LoginID, and 'Chief Executive Officer' for JobTitle. The second row has '2' for BusinessEntityID, 'adventure-works\terri0' for LoginID, and 'Vice President of Engineering' for JobTitle. The third row has '3' for BusinessEntityID, 'adventure-works\roberto0' for LoginID, and 'Engineering Manager' for JobTitle. The fourth row has '4' for BusinessEntityID, 'adventure-works\rob0' for LoginID, and 'Senior Tool Designer' for JobTitle. The fifth row has '5' for BusinessEntityID, 'adventure-works\lgail0' for LoginID, and 'Design Engineer' for JobTitle. The sixth row has '6' for BusinessEntityID, 'adventure-works\jossef0' for LoginID, and 'Design Engineer' for JobTitle. The seventh row has '7' for BusinessEntityID, 'adventure-works\ldylan0' for LoginID, and 'Research and Development Manager' for JobTitle. The eighth row has '10' for BusinessEntityID, 'adventure-works\micha...' for LoginID, and 'Research and Development Manager' for JobTitle.

	BusinessEntityID	LoginID	JobTitle
1	1	adventure-works\ken0	Chief Executive Officer
2	2	adventure-works\terri0	Vice President of Engineering
3	3	adventure-works\roberto0	Engineering Manager
4	4	adventure-works\rob0	Senior Tool Designer
5	5	adventure-works\lgail0	Design Engineer
6	6	adventure-works\jossef0	Design Engineer
7	7	adventure-works\ldylan0	Research and Development Manager
8	10	adventure-works\micha...	Research and Development Manager

6) The HumanResources.Employee table does not contain employee names. Join that table to the Person.Person table on the BusinessEntityID column. Display the job title, birth date, first name, and last name.

```
SELECT
```

```

    e.JobTitle,
    e.BirthDate,
    p.FirstName,
    p.LastName
FROM HumanResources.Employee e
INNER JOIN Person.Person p
ON e.BusinessEntityID = p.BusinessEntityID;

```

Output

100 %

✔ No issues found

Results

Messages

	JobTitle	BirthDate	FirstName	LastName
1	Marketing Assistant	1987-05-03	Kevin	Brown
2	Marketing Specialist	1978-03-06	John	Wood
3	Marketing Assistant	1978-01-29	Mary	Dempsey
4	Marketing Assistant	1975-03-17	Wanida	Benshoof
5	Marketing Specialist	1986-02-04	Terry	Eminhizer
6	Marketing Specialist	1987-05-21	Sariya	Harnpadoungsataya
7	Marketing Specialist	1962-09-13	Mary	Gibson
8	Marketing Specialist	1979-06-18	Jill	Williams

7) Join the Sales.Customer table to the Person.Person table. Display the CustomerID, StoreID, and TerritoryID columns along with the customer name columns.

```

SELECT

```

```

    c.CustomerID,
    c.StoreID,
    c.TerritoryID,

```

p.FirstName,

p.LastName

FROM Sales.Customer c

INNER JOIN Person.Person p

ON c.PersonID = p.BusinessEntityID;

Output

100 %

▼

✔

No issues found

Results

Messages

	CustomerID	StoreID	TerritoryID	FirstName	LastName
1	29487	298	2	Humberto	Acevedo
2	18504	NULL	4	Angel	Adams
3	13280	NULL	1	Bailey	Adams
4	18075	NULL	4	Courtney	Adams
5	25292	NULL	4	Eduardo	Adams
6	12154	NULL	6	Hailey	Adams
7	18987	NULL	1	Hunter	Adams
8	28360	NULL	4	Isaiah	Adams

8) Extend the query written in Question 2 to include the Sales.SalesOrderHeader table. Display the SalesOrderID column along with the columns already specified.

SELECT

c.CustomerID,

```

p.LastName,
p.FirstName,
s.Name AS CompanyName,
soh.SalesOrderID
FROM Sales.Customer c
LEFT JOIN Person.Person p
ON c.PersonID = p.BusinessEntityID
LEFT JOIN Sales.Store s
ON c.StoreID = s.BusinessEntityID
LEFT JOIN Sales.SalesOrderHeader soh
ON c.CustomerID = soh.CustomerID;

```

output

100 % No issues found					
Results Messages					
	CustomerID	LastName	FirstName	CompanyName	SalesOrderID
1	29485	Abel	Catherine	Professional Sales and Service	53459
2	29485	Abel	Catherine	Professional Sales and Service	58907
3	29485	Abel	Catherine	Professional Sales and Service	65157
4	29485	Abel	Catherine	Professional Sales and Service	71782
5	29486	Abercrombie	Kim	Riders Company	49465
6	29486	Abercrombie	Kim	Riders Company	50696
7	29486	Abercrombie	Kim	Riders Company	53526
8	29486	Abercrombie	Kim	Riders Company	58970

9) Write a query that displays all the products along with the SalesOrderID even if an order has never been

placed for that product. Join to the Sales.SalesOrderDetail table using the ProductID column.

```
SELECT  
  
    p.ProductID,  
  
    p.Name,  
  
    sod.SalesOrderID  
  
FROM Production.Product p  
  
LEFT JOIN Sales.SalesOrderDetail sod  
  
ON p.ProductID = sod.ProductID;
```

output

100 %		No issues found	
Results		Messages	
	ProductID	Name	SalesOrderID
1	1	Adjustable Race	NULL
2	2	Bearing Ball	NULL
3	3	BB Ball Bearing	NULL
4	4	Headset Ball Bearings	NULL
5	316	Blade	NULL
6	317	LL Crankarm	NULL
7	318	ML Crankarm	NULL
8	319	HL Crankarm	NULL

10) Write a query that displays the BusinessEntityID, FirstName, MiddleName, and LastName from the Person.Person table for only those rows that have E or B stored in the MiddleName column.

```
SELECT  
  
    BusinessEntityID,  
  
    FirstName,  
  
    MiddleName,  
  
    LastName  
  
FROM Person.Person  
  
WHERE MiddleName IN ('E', 'B');
```

Output

100 % ✓ No issues found				
Results Messages				
	BusinessEntityID	FirstName	MiddleName	LastName
1	285	Syed	E	Abbas
2	38	Kim	B	Abercrombie
3	211	Hazem	E	Abolrous
4	16867	Aaron	B	Adams
5	16845	Kyle	E	Adams
6	16662	Noah	B	Adams
7	287	Amy	E	Alberts
8	13757	Eduardo	E	Alexander